

Zheng Dong

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Education

UC San Diego – M.S. Electrical and Computer Engineering

Sep 2025 – Jun 2027

Pennsylvania State University – B.S. Data Science

Sep 2021 – May 2025

Research Interests

Reliable **LLM** systems, **LLM agents**, **retrieval-augmented generation**, **trustworthy ML**, **data-centric AI**, **NLP information extraction**, **ML systems**, and **AI for scientific/high-stakes decision support**.

Experience

Permitfolio

Jun 2026 – Present

Machine Learning Engineer Intern

New York, United States

- Built auditable **LLM**-assisted asset-classification service with **Python**, **FastAPI**, **Pydantic**, and **PostgreSQL**, parsing raw holding descriptions and mapping them to regulatory asset classes at **92%** accuracy across 20+ labeled test fixtures
- Implemented reliability layer with per-prediction **confidence calibration** and threshold-based human-in-the-loop routing, sending low-confidence cases to manual audit and cutting reviewer workload by **45%** while preserving full model-version traceability
- Created **pytest** unit and integration suite reaching **90%** coverage and automated **CI/CD** with **GitHub Actions** and **Docker**, enabling reproducible model-service builds and consistent staging-to-production deploys

Penn State University

Jan 2025 – May 2025

Machine Learning Research Assistant

University Park, PA

- Implemented **NLP** information-extraction pipeline with **spaCy** and **BERT** fine-tuning over domain-specific preprocessing, processing **10K+** documents/day through **Kafka** streaming and reaching **92%** F1 on held-out evaluation
- Developed reproducible experiment-tracking workflow with **MLflow** and **Optuna** hyperparameter search, logging interpretability and fairness metrics per run to surface bias early across model iterations
- Deployed entity-extraction microservice in **Java** and **Python** with **CI/CD** and high test coverage, resolving **N+1** query patterns and adding index tuning that cut request latency by **18%**

Allianz Insurance

Jun 2024 – Sep 2024

Machine Learning Engineer Intern

Qingdao, China

- Built end-to-end fraud-detection **ML** pipeline with **Python** and **PySpark** over **200K+** daily transactions, engineering privacy-preserving aggregate features that improved risk-scoring precision by **15%**
- Developed real-time model-serving API in **Flask** with **Redis** write-through caching and connection pooling, reducing p95 inference latency by **30%** (**150ms** to **105ms**) under concurrent load
- Deployed gradient-boosting model with **MLflow** and **Docker**, adding automated fairness monitoring that tracked demographic parity and equalized odds across protected groups in production

Qingdao Engineering Consulting Institute

Jun 2023 – Sep 2023

Software Engineer Intern

Qingdao, China

- Developed real-time monitoring platform on **Linux** streaming **500+** sensor feeds over **WebSocket**, adding circuit-breaker fault tolerance and graceful degradation that raised system uptime from **97%** to **99.5%**
- Implemented anomaly-detection service over **100K+** daily sensor readings using time-series feature engineering and sliding-window aggregation, achieving **79%** precision at sub- **2%** false-positive rate

Projects

Reliable LLM Copilot for Portfolio Risk (mindmarket.app)

Apr 2026 – Present

- Built full-stack AI system with **Next.js**, **TypeScript**, and **FastAPI**, grounding **Claude** agent on computed evidence through **MCP** tool surface so generated answers cite deterministic outputs instead of fabricating numbers
- Developed deterministic quant engine in **NumPy**, **pandas**, and **SciPy** computing **VaR**, **CVaR**, factor betas, and stress scenarios that serve as verifiable evidence for agent reasoning
- Deployed production stack to **AWS EC2** with **Docker Compose**, **Caddy**, and **GitHub Actions** CI/CD, maintaining **460+** **pytest** tests alongside **Stripe** billing and **Sentry** / **PostHog** observability

Multimodal Retrieval-Augmented Generation (RAG) System

Jul 2025 – Oct 2025

- Built multimodal retrieval combining **CLIP** and **OpenAI** embeddings with rank-fusion reranking, deploying **Pinecone** vector search (chosen over **Chroma** for managed scaling at **50K+** items) at **92% Recall@5** on **1K** eval set
- Developed **FastAPI** backend serving **7** REST endpoints with input validation and error handling, validated by **JMeter** load testing at sub- **1s** p95 latency under **500** concurrent requests

Distributed LLM Inference System – Kaggle Silver Medal (Top 5%)

Jun 2024 – Aug 2024

- Optimized distributed multi-GPU **PyTorch** inference with dynamic padding and **ThreadPoolExecutor** parallelism, achieving **3x** throughput gain and **65%** latency reduction validated over **20K+** daily batch predictions

Skills

ML & LLM: PyTorch, Transformers, BERT, LLMs, RAG, LLM Agents, NLP, Deep Learning, Embeddings, CLIP

ML Systems & MLOps: MLflow, TorchServe, Optuna, PySpark, Feature Engineering, Evaluation, Fairness Monitoring

Languages: Python, Java, C++, SQL, TypeScript

Tools & Cloud: FastAPI, Docker, AWS (EC2, Lambda, S3), Kafka, Redis, PostgreSQL, CI/CD, GitHub Actions, Linux